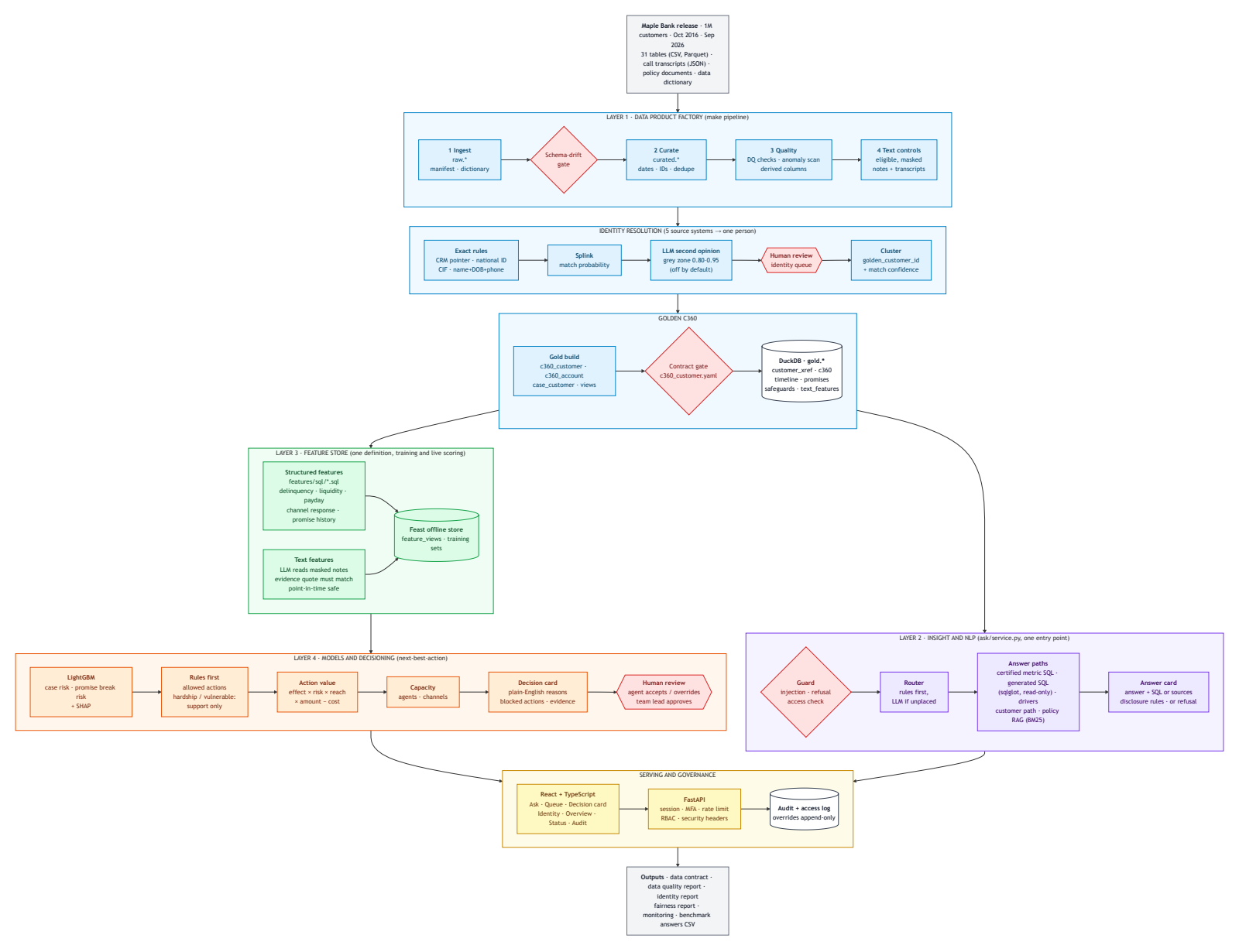

Maple Bank release · 1M customers · Oct 2016 · Sep 2026
31 tables (CSV, Parquet) · call transcripts (JSON) · policy documents · data dictionary
LAYER 1 · DATA PRODUCT FACTORY (make pipeline)
1 Ingest
raw.* manifest · dictionary
Schema-drift gate
2 Curate
curated.* dates · IDs · dedupe
3 Quality
DQ checks · anomaly scan derived columns
4 Text controls
eligible, masked notes + transcripts
IDENTITY RESOLUTION (5 source systems → one person)
Exact rules
CRM pointer · national ID CIF · name+DOB+phone
Splink
match probability
LLM second opinion
grey zone 0.80–0.95 (off by default)
Human review
identity queue
Cluster
golden_customer_id + match confidence
GOLDEN C360
Gold build
c360_customer · c360_account case_customer · views
Contract gate
c360_customer.yaml
DuckDB · gold.*
customer_xref · c360 timeline · promises safeguards · text_features
LAYER 3 · FEATURE STORE (one definition, training and live scoring)
Structured features
features/sql/*.sql
delinquency · liquidity · payday
channel response · promise history
Text features
LLM reads masked notes evidence quote must match point-in-time safe
Feast offline store
feature_views · training sets
LAYER 2 · INSIGHT AND NLP (ask/service.py, one entry point)
Guard
injection · refusal access check
Router
rules first, LLM if unplaced
Answer paths
certified metric SQL · generated SQL (sqlglot, read-only) · drivers
customer path · policy RAG (BM25)
Answer card
answer + SQL or sources disclosure rules · or refusal
LAYER 4 · MODELS AND DECISIONING (next-best-action)
LightGBM
case risk · promise break risk + SHAP
Rules first
allowed actions hardship / vulnerable: support only
Action value
effect × risk × reach × amount − cost
Capacity
agents · channels
Decision card
plain-English reasons blocked actions · evidence
Human review
agent accepts / overrides team lead approves
SERVING AND GOVERNANCE
React + TypeScript
Ask · Queue · Decision card Identity · Overview · Status · Audit
FastAPI
session · MFA · rate limit RBAC · security headers
Audit + access log
overrides append-only
Outputs · data contract · data quality report · identity report fairness report · monitoring · benchmark answers CSV